
MY YOUTUBE CHANNEL S.KHAN ACADEMY
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KHAN

MTH101 QUIZ NO 3 SPRING 2021 || 100% SOLVED ANSWER

Quiz Start Time: 09:30 AM, 14 July 2021

Question # 1 of 10 (Start time: 09:30:36 AM, 14 July 2021)

Total Marks: 1

Why the equation: x^2 + 8 = 0 does not have approximate solution while using Newton's method?

Select the correct option

- x^2 will always be negative
- x^2 will always be nonnegative

Click to Save Answer & Move to Next Question

S KHAN ACADEMY

0:04 / 9:18

MTH101:Quiz 2

Quiz Start Time: 09:30 AM, 14 July 2021

Question # 2 of 10 (Start time: 09:32:21 AM, 14 July 2021)

Total Marks: 1

What are critical points of the function f(x) =x-1?

Select the correct option

- ○ No critical point
- ○ None of these
- ● x=1
- ○ x=0

Click to Save Answer & Move to Next Question

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MTH101:Quiz 2

Quiz Start Time: 09:30 AM, 14 July 2021

Question # 3 of 10 (Start time: 09:33:54 AM, 14 July 2021)

Total Marks: 1

Let A be the area of a rectangle under a continuous function f(x) over a closed interval [a, b]. If this area is divided in to 'n' sub-rectangles then width of each approximated sub-intervals is ---------

Select the correct option

- (a-b)/2
- (a-b)/n
- (b-a)/2n
- (b-a)/n

Click to Save Answer & Move to Next Question

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MTH101:Quiz 2

Quiz Start Time: 09:30 AM, 14 July 2021

Question # 4 of 10 (Start time: 09:35:34 AM, 14 July 2021)

Total Marks: 1

Increase in number of rectangles under any continuous function gives ..... approximation to area.

Select the correct option

- better
- Poor
- no change in
- None of these

Click to Save Answer & Move to Next Question

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MTH101:Quiz 2

Quiz Start Time: 09:30 AM, 14 July 2021

Question # 5 of 10 (Start time: 09:36:39 AM, 14 July 2021)

Total Marks: 1

If Newton's Method succeeded to get the approximate solution of an equation, then which of the following is NOT true about it.

Select the correct option

- ○ The slope of the tangent line (at any approximated point) must be non zero.
- ○ None of these.
- ● The tangent line (at any approximated point) is not parallel to x-axis.
- ○ The sequence of approximated points not convergent to the exact solution

Click to Save Answer & Move to Next Question

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MTH101:Quiz 2

Quiz Start Time: 09:30 AM, 14 July 2021

Question # 6 of 10 (Start time: 09:38:26 AM, 14 July 2021)

Total Marks: 1

Summation of 2 where sum ranges from 0 to 10 equals 20.

Select the correct option

- ○ False
- ● True

Click to Save Answer & Move to Next Question

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MTH101:Quiz 2

Quiz Start Time: 09:30 AM, 14 July 2021

Question # 7 of 10 (Start time: 09:39:59 AM, 14 July 2021)

Total Marks: 1

If the closed interval [-10,x] is divided into '20' equally spaced subintervals each of which having the width equals to '1' unit then the value of 'x' is --------

Select the correct option

- 0
- 10
- 30
- 20

Click to Save Answer & Move to Next Question

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MTH101:Quiz 2

Quiz Start Time: 09:30 AM, 14 July 2021

Question # 8 of 10 (Start time: 09:41:29 AM, 14 July 2021)

Total Marks: 1

If f(x) = x^4, then which of the following is Not true about it.

Select the correct option

- (●) Its anti - derivative is x^5/5.
- ( ) Its anti - derivative is x^5/5 + 10.
- ( ) Its anti - derivative is x^5 + 5.
- ( ) Its anti - derivative is x^5/5 + 2.

Click to Save Answer & Move to Next Question

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MTH101:Quiz 2 Quiz Start Time: 09:30 AM, 14 July 2021

Question # 9 of 10 (Start time: 09:42:27 AM, 14 July 2021) Total Marks: 1

Let y = f(x) be a discontinuous function on a finite closed interval, then which of the following is true about it.

Select the correct option

- It has only absolute minimum value.
- None of these.
- It may or may not have absolute extreme values.
- It must have absolute extreme values.

Click to Save Answer & Move to Next Question

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MTH101:Quiz 2

Quiz Start Time: 09:46 AM, 14 July 2021

Question # 1 of 10 (Start time: 09:47:09 AM, 14 July 2021)

Total Marks: 1

Integral of 5^2 is

NOTE: x^n means 'x' to the power 'n'

Select the correct option

- 25x
- (1/3)5^3
- None of these
- 10

Click to Save Answer & Move to Next Question

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MTH101 QUIZ NO 3 SPRING 2021 || 100% SOLVED ANSWER

MTH101 Quiz 2

Quiz Start Time: 09:30 AM, 14 July 2021

Question # 10 of 10 (Start time: 09:44:04 AM, 14 July 2021) **Total Marks: 1**

If f (x) = x^3 is defined on the interval [1, 3], then which of the following is true about it.

Select the correct option

- ○ Its relative minimum value exists at the critical point.
- ● Its relative minimum value does not exist at the critical point
- ○ None of these.
- ○ Its absolute minimum value exists at the critical point.

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Click to Save Answer & Move to Next Question

1:39 / 9:18

MTH101:Quiz 2

Quiz Start Time: 09:46 AM, 14 July 2021

Question # 1 of 10 (Start time: 09:47:09 AM, 14 July 2021)

Total Marks: 1

Integral of 5^2 is

NOTE: x^n means 'x' to the power 'n'

Select the correct option

- 25x
- (1/3)5^3
- None of these
- 10

Click to Save Answer & Move to Next Question

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MTH101:Quiz 2

Quiz Start Time: 09:46 AM, 14 July 2021

Question # 2 of 10 (Start time: 09:47:40 AM, 14 July 2021)

Total Marks: 1

In approximation to an area Rn (where n is subscript) when limit is taken as n goes to infinity, approximation becomes actual area.

Select the correct option

- False
- True

Click to Save Answer & Move to Next Question

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MTH101:Quiz 2

Quiz Start Time: 09:46 AM, 14 July 2021

Question # 3 of 10 (Start time: 09:49:16 AM, 14 July 2021)

Total Marks: 1

If f(x) = Sec^ (2) x + x^3, then which of the following is NOT true about it.

Select the correct option

- Its anti - derivative is Tan(x) + x^4/4 + 15.
- None of these.
- Its anti - derivative is Tan(x) + x^4/4 + 12.
- Its anti - derivative is Tan(x) + x^4/4 + 10.

Click to Save Answer & Move to Next Question

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MTH101 Quiz 2

Quiz Start Time: 09:46 AM, 14 July 2021

Question # 6 of 10 (Start time: 09:52:09 AM, 14 July 2021)

Total Marks: 1

Integration of 5 with respect to x is.....

Select the correct option

- 5
- x
- 5x^2
- 5x

Click to Save Answer & Move to Next Question

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2:33 / 9:18

MTH101:Quiz 2

Quiz Start Time: 09:46 AM, 14 July 2021

Question # 6 of 10 (Start time: 09:52:09 AM, 14 July 2021)

Total Marks: 1

Integration of 5 with respect to x is.....

Select the correct option

- ○ 5
- ○ x
- ○ 5x^2
- ● 5x

Saving...

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MTH101 QUIZ NO 3 SPRING 2021 || 100% SOLVED ANSWER

Quiz Start Time: 09:46 AM, 14 July 2021

Question # 8 of 10 (Start time: 09:54:56 AM, 14 July 2021)

Total Marks: 1

Sum of n-terms of a series whose nth term is 'n' = 1/n+1.then what is the sum of the first two terms is -----

Select the correct option

- (●) 5/6
- ( ) 6/5
- ( ) 6/4
- ( ) 6

Saving...

2:51 / 9:18

Double Tap to Edit
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MTH101:Quiz 2

Quiz Start Time: 09:46 AM, 14 July 2021

Question # 9 of 10 (Start time: 09:56:15 AM, 14 July 2021)

Total Marks: 1

Newton's method uses the ..... to approximate the root.

Select the correct option

- ○ None of these
- ● Tangent line
- ○ Secant line
- ○ Normal line

Click to Save Answer & Move to Next Question

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MTH101:Quiz 2

Quiz Start Time: 10:02 AM, 14 July 2021

Question # 1 of 10 (Start time: 10:02:43 AM, 14 July 2021)

Total Marks: 1

The polynomial function f(x)=6x^2-30x+36 has the critical point over the real line is .....

Select the correct option

- 5
- 2
- 2/5
- 5/2

Click to Save Answer & Move to Next Question

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MTH101:Quiz 2

Quiz Start Time: 10:02 AM, 14 July 2021

Question # 2 of 10 (Start time: 10:04:20 AM, 14 July 2021)

Total Marks: 1

If f(x) = x^5 + 6, then which of the following is Not true about it.

Select the correct option

- (●) Its anti - derivative is x^6/6 + 6x + 6.
- ( ) Its anti - derivative is x^6/6 + 6x.
- ( ) Its anti - derivative is x^6/6 + 6x + 10.
- ( ) Its anti - derivative is x^6/6 + 12x

Click to Save Answer & Move to Next Question

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MTH101 QUIZ NO 3 SPRING 2021 || 100% SOLVED ANSWER

Quiz Start Time: 10:02 AM, 14 July 2021

Question # 3 of 10 (Start time: 10:05:00 AM, 14 July 2021)

Total Marks: 1

If x = 1 + 2 +3 + 4 + . . . + 20, then
x = _____.

Select the correct option

- None of these.
- 200.
- 500.
- 210.

Click to Save Answer & Move to Next Question

3:25 / 9:18

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MTH101:Quiz 2

Quiz Start Time: 10:02 AM, 14 July 2021

Question # 4 of 10 (Start time: 10:05:42 AM, 14 July 2021)

Total Marks: 1

In sigma notation 12+14+16+18+20 can be written as.....

Select the correct option

- summation of (k) where (k varies from 6 to10)
- summation of (2k) where (k varies from 1 to5)
- summation of (k^2) where (k varies from 6 to10)
- summation of (2k) where (k varies from 6 to10)

Click to Save Answer & Move to Next Question

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MTH101 Quiz 2

Quiz Start Time: 10:02 AM, 14 July 2021

Question # 7 of 10 (Start time: 10:09:22 AM, 14 July 2021)

Total Marks: 1

x=1 is a critical value of the function.
f(x)=(x-1)^3

NOTE: x^n means 'x' to the power 'n'

Select the correct option

- False
- True

Click to Save Answer & Move to Next Question

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3:55 / 9:18

MTH101:Quiz 2 Quiz Start Time: 10:02 AM, 14 July 2021

Question # 10 of 10 (Start time: 10:13:05 AM, 14 July 2021) Total Marks: 1

If f (x) = 2 x + 7 is defined on the interval
[2, 4), then which of the following is true about it.

Select the correct option

- ○ It has only absolute minimum value.
- ○ None of these.
- ● It has both absolute maximum and minimum values.
- ○ It has only absolute maximum value.

Click to Save Answer & Move to Next Question

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MTH101 Quiz 2

Quiz Start Time: 10:14 AM, 14 July 2021

Question # 1 of 10 (Start time: 10:14:46 AM, 14 July 2021)

Total Marks: 1

The indefinite integral of 5sinx is .....

Select the correct option

- 5cosx +c
- -5cosx+c
- cosx/5+c
- -cosx/5+c

Click to Save Answer & Move to Next Question

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4:21 / 9:18

MTH101 QUIZ NO 3 SPRING 2021 || 100% SOLVED ANSWER

MTH101:Quiz 2

Quiz Start Time: 10:14 AM, 14 July 2021

Question # 2 of 10 (Start time: 10:15:00 AM, 14 July 2021)

Total Marks: 1

summation of (ai) (i varies from 1 to n) , summation of (aj) (j varies from 1 to n),summation of (ak) (k varies from 1 to n)
All these three represents same summation.

Select the correct option

- True
- False

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Click to Save Answer & Move to Next Question

4:25 / 9:18

MTH101:Quiz 2

Quiz Start Time: 10:14 AM, 14 July 2021

Question # 2 of 10 (Start time: 10:15:00 AM, 14 July 2021)

Total Marks: 1

summation of (ai) (i varies from 1 to n) , summation of (aj) (j varies from 1 to n),summation of (ak) (k varies from 1 to n)
All these three represents same summation.

Select the correct option

- True
- False

Click to Save Answer & Move to Next Question

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MTH101 QUIZ NO 3 SPRING 2021 || 100% SOLVED ANSWER

MTH101 Quiz 2

Quiz Start Time: 10:26 AM, 14 July 2021

Question # 1 of 10 (Start time: 10:26:46 AM, 14 July 2021)

Total Marks: 1

Which of the following will be left end points if the interval [-2,2] is divided into 4 equal subintervals.

Select the correct option

- ● -2,-1,1,2
- ○ -2,-1,0,1
- ○ None of these
- ○ -1,0,1,2

Click to Save Answer & Move to Next Question

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5:00 / 9:18

MTH101 QUIZ NO 3 SPRING 2021 || 100% SOLVED ANSWER

MTH101 Quiz 2 — Quiz Start Time: 10:26 AM, 14 July 2021

Question # 6 of 10 (Start time: 10:32:46 AM, 14 July 2021) — **Total Marks: 1**

Why the equation: x^2 + 8 = 0 does not have approximate solution while using Newton's method?

Select the correct option

- x^2 will always be negative
- x^2 will always be nonnegative

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MTH101 QUIZ NO 3 SPRING 2021 || 100% SOLVED ANSWER

MTH101 Quiz 2

Quiz Start Time: 10:26 AM, 14 July 2021

Question # 9 of 10 (Start time: 10:33:23 AM, 14 July 2021)

Total Marks: 1

If x = (1^2)+(2^2)+(3^2)+(4^2) + . . . + (30^2), then x = _____.

Select the correct option

- 900.
- None of these.
- 9455.
- 465.

Click to Save Answer & Move to Next Question

5:56 / 9:18

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MTH101 QUIZ NO 3 SPRING 2021 || 100% SOLVED ANSWER

MTH101 Quiz 2

Quiz Start Time: 10:26 AM, 14 July 2021

Question # 10 of 10 (Start time: 10:34:02 AM, 14 July 2021)

Total Marks: 1

If f(x)=Tan(x) then mean value theorem can be applied to it on the interval (0,2pi)

Select the correct option

- ○ True
- ● False

Click to Save Answer & Move to Next Question

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6:02 / 9:18

MTH101 QUIZ NO 3 SPRING 2021 || 100% SOLVED ANSWER

MTH101 Quiz 2 — Quiz Start Time: 10:36 AM, 14 July 2021

Question # 2 of 10 (Start time: 10:36:16 AM, 14 July 2021) — Total Marks: 1

If 'n' goes from 1 to 4 and the summation of 'na' =Maxima of (e^x) in the interval[-e,0], then the value of 'a' is ----------

Select the correct option

- -1/10
- 1/10
- 10
- -10

Click to Save Answer & Move to Next Question

6:08 / 9:18

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MTH101 Quiz 2

Quiz Start Time: 10:36 AM, 14 July 2021

Question # 3 of 10 (Start time: 10:37:46 AM, 14 July 2021)

Total Marks: 1

Subdivide the interval [a, b] into 4 equal subintervals then the width of each subinterval is ------

Select the correct option

- (b-a)/2
- (b-2a)/4
- (b-a)/4
- (2b-a)/4

Click to Save Answer & Move to Next Question

6:13 / 9:18

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MTH101 QUIZ NO 3 SPRING 2021 || 100% SOLVED ANSWER

MTH101 Quiz 2

Quiz Start Time: 10:36 AM, 14 July 2021

Question # 6 of 10 (Start time: 10:38:55 AM, 14 July 2021)

Total Marks: 1

If f(x) = x^4, then which of the following is Not true about it.

Select the correct option

- (●) Its anti - derivative is x^5 + 5.
- ( ) Its anti - derivative is x^5/5.
- ( ) Its anti - derivative is x^5/5 + 10.
- ( ) Its anti - derivative is x^5/5 + 2.

Click to Save Answer & Move to Next Question

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6:30 / 9:18

MTH101 QUIZ NO 3 SPRING 2021 || 100% SOLVED ANSWER

Quiz Start Time: 10:36 AM, 14 July 2021

Question # 7 of 10 (Start time: 10:39:21 AM, 14 July 2021)

Total Marks: 1

Subdivide the interval [3, 5] into n equal parts, and then the width of each subinterval is ----

Select the correct option

- 1/n
- n
- 2/n
- -2/n

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Click to Save Answer & Move to Next Question

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MTH101:Quiz 2

Quiz Start Time: 10:36 AM, 14 July 2021

Question # 7 of 10 (Start time: 10:39:21 AM, 14 July 2021)

Total Marks: 1

Subdivide the interval [3, 5] into n equal parts, and then the width of each subinterval is ----

Select the correct option

- ○ 1/n
- ○ n
- ● 2/n
- ○ -2/n

Click to Save Answer & Move to Next Question

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MTH101 QUIZ NO 3 SPRING 2021 || 100% SOLVED ANSWER

MTH101 Quiz 2

Quiz Start Time: 10:36 AM, 14 July 2021

Question # 8 of 10 (Start time: 10:39:42 AM, 14 July 2021)

Total Marks: 1

If f(x) = x^5 + x, then which of the following is true about it.

Select the correct option

- Its anti - derivative is x^6/6 + x^2/2 + 6.
- Its anti - derivative is x^5/5 + 1.
- Its anti - derivative is 5x^4 + 1.
- None of these.

Click to Save Answer & Move to Next Question

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7:05 / 9:18

MTH101:Quiz 2 Quiz Start Time: 10:36 AM, 14 July 2021

Question # 9 of 10 (Start time: 10:40:46 AM, 14 July 2021) Total Marks: 1

Newton's Method fails to find the approximate solution of an equation if _____.

Select the correct option

○	the tangent line(at any approximated point) is parallel to x-axis.
○	None of these
○	the slope of the tangent line(at any approximated point) is non-zero
●	the tangent line (at any approximated point) is not parallel to x-axis.

Saving...

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MTH101:Quiz 2

Quiz Start Time: 10:36 AM, 14 July 2021

Question # 10 of 10 (Start time: 10:41:45 AM, 14 July 2021)

Total Marks: 1

If 'n' goes from 1 to 3 and the summation of 'na' = 6a, then the value of 'a' is ----------

Select the correct option

- Undetermined
- 1
- -6
- 6

Click to Save Answer & Move to Next Question

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MTH101:Quiz 2

Quiz Start Time: 10:42 AM, 14 July 2021

Question # 3 of 10 (Start time: 10:44:46 AM, 14 July 2021)

Total Marks: 1

1+2+3.....+1000 equals -------

Select the correct option

- None of these
- 3000
- 1000
- 500500

Click to Save Answer & Move to Next Question

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MTH101 QUIZ NO 3 SPRING 2021 || 100% SOLVED ANSWER

Quiz Start Time: 10:42 AM, 14 July 2021

Question # 4 of 10 (Start time: 10:46:19 AM, 14 July 2021) **Total Marks: 1**

If f (x) = |x| -2 is defined on the interval
[-2, 2], then which of the following is true about it.

Select the correct option

- None of these.
- There is a point in the interval (-2, 2) where f(x) has a horizontal tangent
- There is no such point in the interval (-2, 2) where f(x) has a horizontal tangent.
- It is discontinuous on the interval [-2, 2].

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Click to Save Answer & Move to Next Question

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MTH101 QUIZ NO 3 SPRING 2021 || 100% SOLVED ANSWER

Quiz Start Time: 10:42 AM, 14 July 2021

Question # 5 of 10 (Start time: 10:46:51 AM, 14 July 2021)

Total Marks: 1

Which of the following is the absolute minima of the function f(x)=-x in the interval [-1,1]?

Select the correct option

- (●) -1
- ( ) 1
- ( ) 0.5
- ( ) 0

Click to Save Answer & Move to Next Question

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Search the web and Windows

ENG 4:54 PM 7/14/2021

MTH101 QUIZ NO 3 SPRING 2021 || 100% SOLVED ANSWER

Quiz Start Time: 10:42 AM, 14 July 2021

Question # 5 of 10 (Start time: 10:46:51 AM, 14 July 2021) **Total Marks: 1**

Which of the following is the absolute minima of the function: f(x)=-x in the interval [-1,1]?

Select the correct option

- ○ -1
- ● 1
- ○ 0.5
- ○ 0

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Saving...

Search the web and Windows

ENG 4:54 PM 7/14/2021

MTH101:Quiz 2

Quiz Start Time: 10:42 AM, 14 July 2021

Question # 6 of 10 (Start time: 10:47:55 AM, 14 July 2021)

Total Marks: 1

If f(x) = Cos(x) + Sin(x) + x, then which of the following is NOT true about it.

Select the correct option

- Its anti - derivative is -Sin(x) - Cos(x) + x^2/2 + 4.
- None of these.
- Its anti - derivative is Sin(x) - Cos(x) + x^2/2 + 4.
- Its anti - derivative is Sin(x) - Cos(x) + x^2/2 + 5.

Click to Save Answer & Move to Next Question

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MTH101:Quiz 2

Quiz Start Time: 10:42 AM, 14 July 2021

Question # 8 of 10 (Start time: 10:50:02 AM, 14 July 2021)

Total Marks: 1

Maximum of the function f(x)=2x+7 occurs at

Select the correct option

- None of these
- x=-7/2
- x=-2/7
- x=0

Click to Save Answer & Move to Next Question

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MTH101:Quiz 2

Quiz Start Time: 10:42 AM, 14 July 2021

Question # 8 of 10 (Start time: 10:50:02 AM, 14 July 2021)

Total Marks: 1

Maximum of the function f(x)=2x+7 occurs at

Select the correct option

- None of these
- x=-7/2
- x=-2/7
- x=0

Click to Save Answer & Move to Next Question

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MTH101:Quiz 2

Quiz Start Time: 10:42 AM, 14 July 2021

Question # 9 of 10 (Start time: 10:50:26 AM, 14 July 2021)

Total Marks: 1

What are critical points of the function f(x) =x^2-1?
NOTE: x^n means 'x' to the power 'n'

Select the correct option

- ○ No critical point
- ● x=0
- ○ None of these
- ○ x=1

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